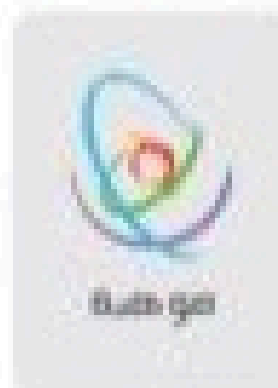


Reducing Carbon Emissions of Factories by Using Artificial Intelligence

مؤسسة الملك عبدالعزيز ورجاله للموهبة والإبداع
King Abdulaziz & His Companions Foundation for Giftedness & Creativity



Emad Ali Alharbi. Supervised by: Dr. Abdulrahman Alofi.
Department/Center: Intelligent Manufacturing and Robotics (IMR)
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Introduction:

Factories are one of the biggest producers of carbon emissions. They contribute to a fifth of the world's total carbon emissions, and their emissions are steadily increasing; reducing these carbon emissions is of great importance. Factories contain various components, with each of them having a variety of types, each with its own advantages and disadvantages, so identifying which ones you should use proves to be troubling.

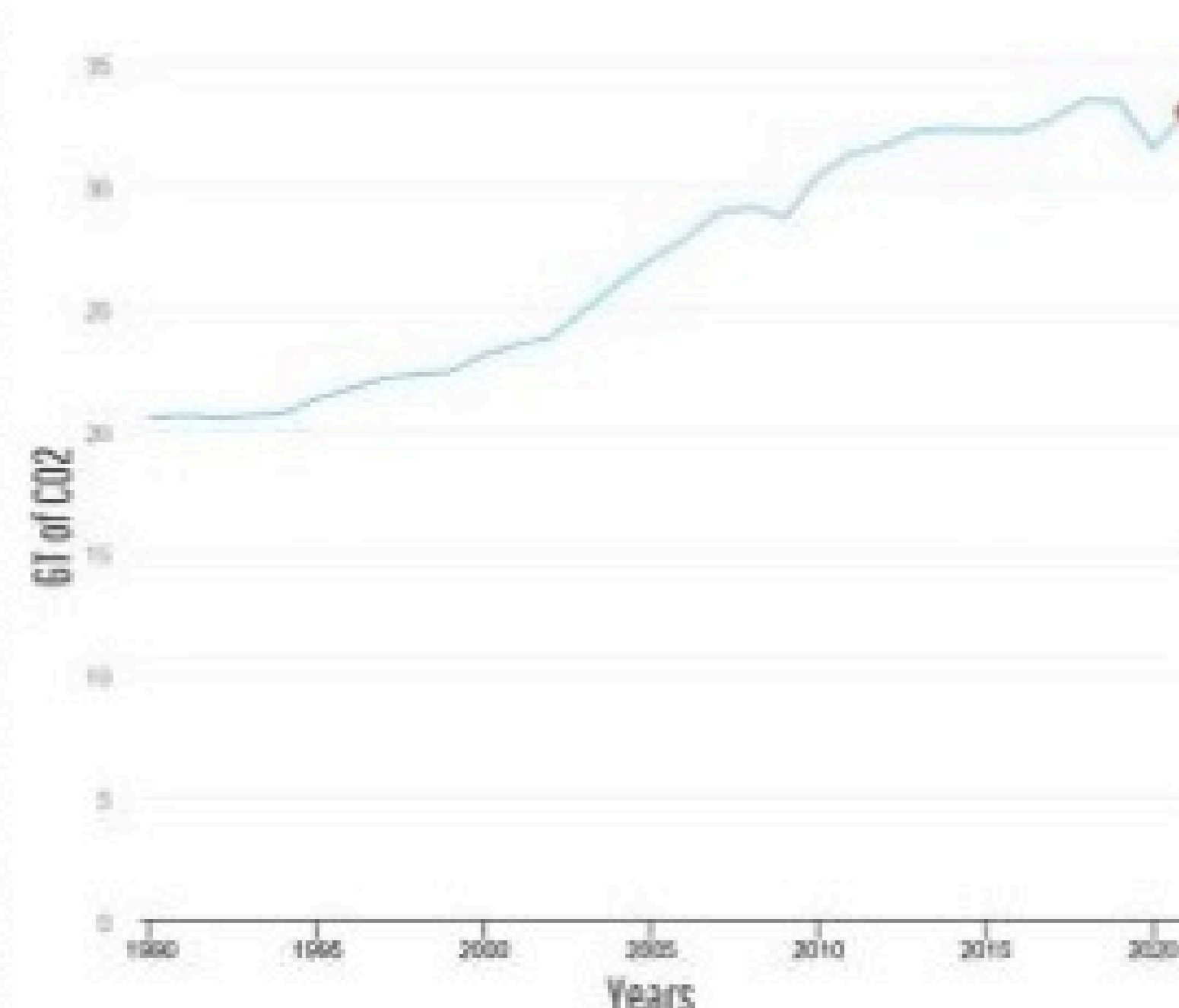


Figure 1: The amount of CO2 in gigatons produced by factories in the years from 1990 to 2020

Research Problem:

The decrease in emissions must come without affecting productivity or increasing costs. The potential role of AI in reducing emissions needs more research to build AI models that can help identify the optimal combination of factory properties and variables to achieve the best results and give accurate rankings of a factory. AI algorithms such as Neural Networks and Genetic algorithms will prove useful to the problem for their ability to handle large inputs quickly and with ease.

Objectives:

- Create an AI model to find the combination of components with the lowest carbon emissions.
- Create an AI Neural network that ranks a factory based on specific inputs.

Factory				
10	5	8	9	8
8	3	6	8	7

Total Rating 7
Eco friendly
Dangerous emissions

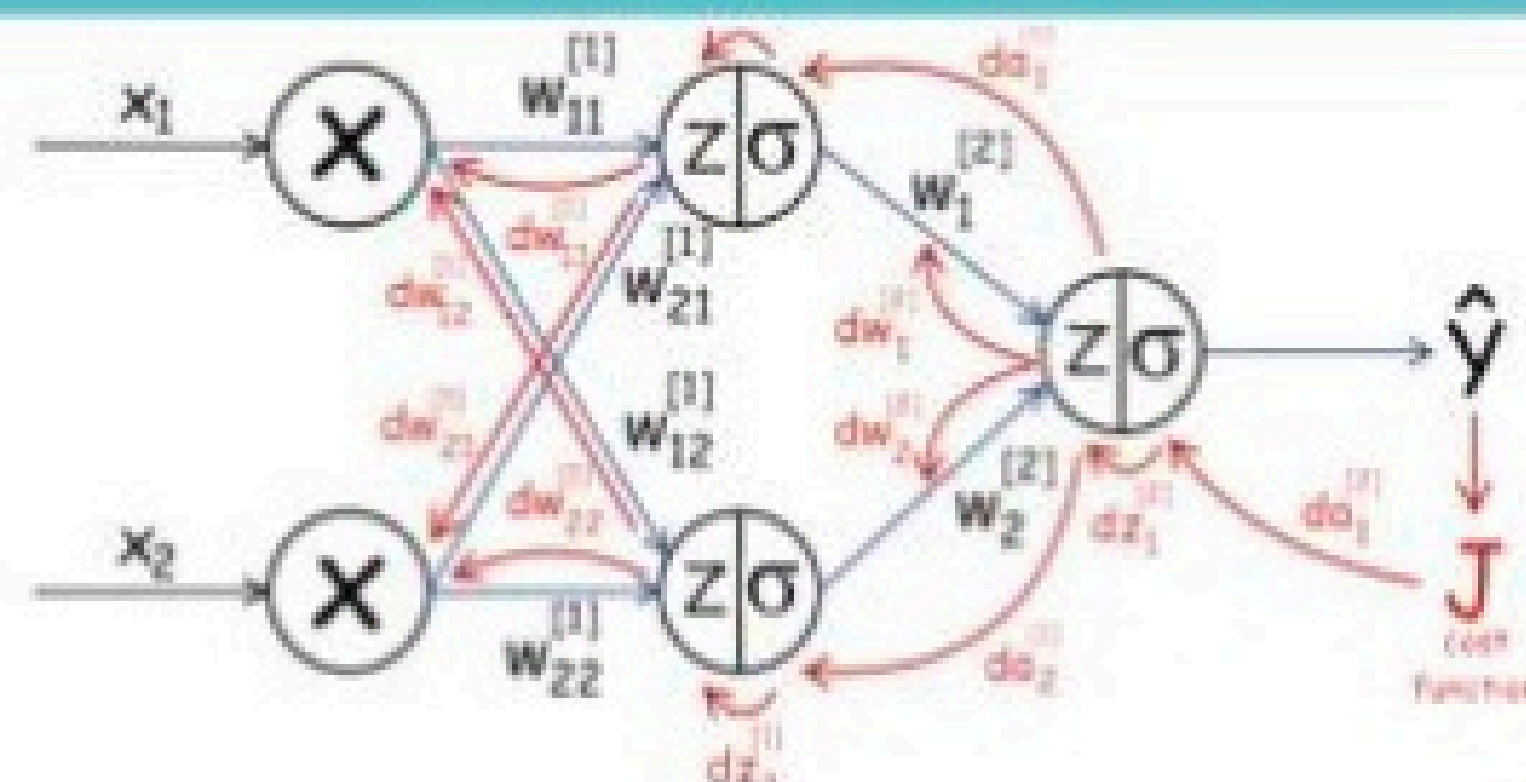
Figure 2: the representation of a factory for the AI models

Methodology:

The methodology consists of two stages:

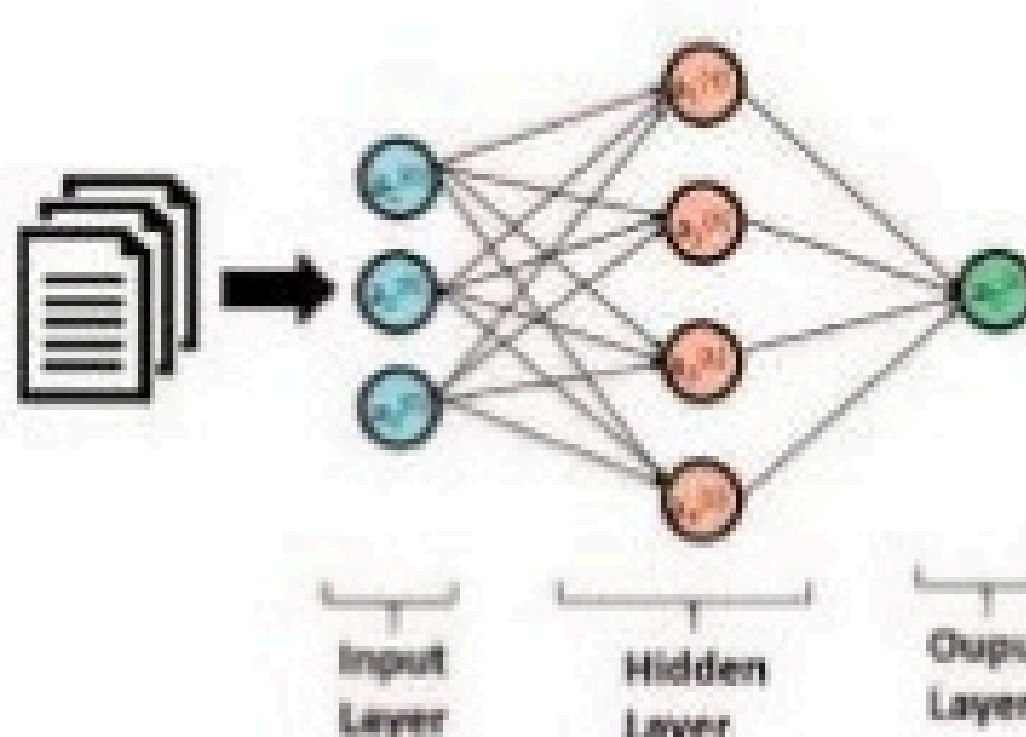
1. Finding the ideal combination with multi-objective optimization using a Genetic algorithm.
2. Ranking a specific factory with a Neural Network

- In order to achieve the best outcomes, the genetic algorithm executes three distinct functions hundreds of times: Evaluation, Reproduction, and Mutation.
- Python 3 was used to implement this algorithm.
- The two distinct functions that make up the neural network are executed hundreds of times each: Forwardpropagation, and Backpropagation.
- Matlab was used to implement this algorithm.



Backpropagation

Figure 3: The second process of the Neural Network: Backpropagation, where the algorithm uses the error from the forward propagation to calculate the effect of each weight on the relative error by calculating the derivative of the error with respect to each weight



Forwardpropagation

Figure 5: The first process of the Neural Network: Forwardpropagation, where the algorithm takes the inputs and runs them through the network using the equation for each neuron to give a final output and compare it to the expected result to give a total error.

Fitness = 2	3	1	6
Fitness = 1	3	5	2

Figure 4: The first process of the Genetic algorithm, Evaluation, where you assign each factory a ranking based on how many other factories are worse than it and then take only the good factories.

Evaluation

6	2	4	3
3	5	1	7

Figure 6: The second process of the Genetic algorithm, Reproduction, where each two factories chosen from the Evaluation will make two new factories, each sharing half of the parents components.

Reproduction

4	7	2
4	3	2

Figure 7: The third process of the Genetic algorithm, Mutation where a small number of factories are taken and random components are randomized

Mutation

Results:

By using Genetic algorithms, this project was able to significantly reduce the carbon emissions of factories and give them a near optimal combination of components without affecting their cost or productivity, while also being extremely fast and able to handle large input data sets of up to 1000 components and 1000 replacements with ease.

With the Neural networks, I was able to accurately give a ranking from 1 to 100 with minor errors.

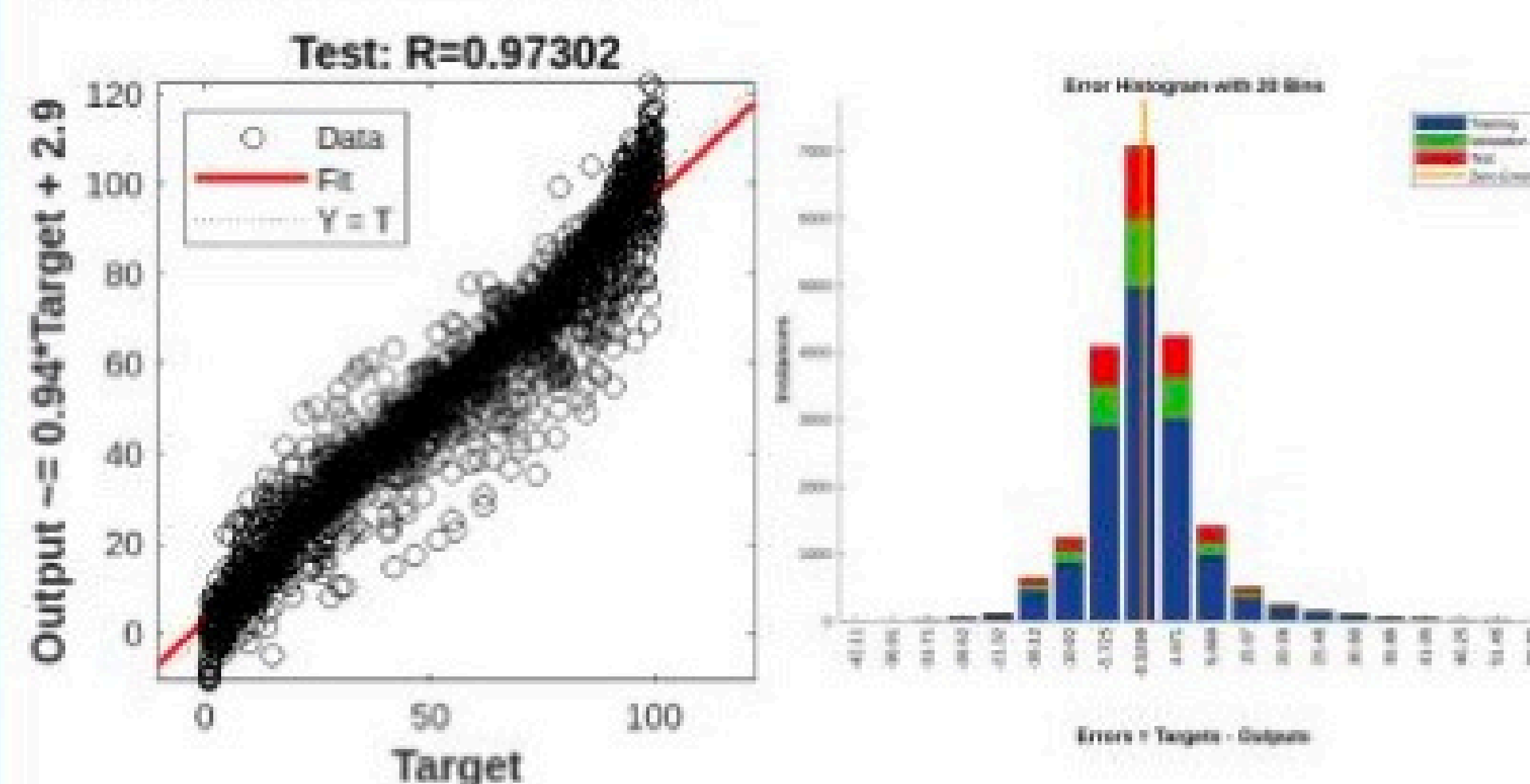


Figure 8: The total regression of the test run on the Neural Network. The red line is the expected result, and the black circles are the results given from a certain input.

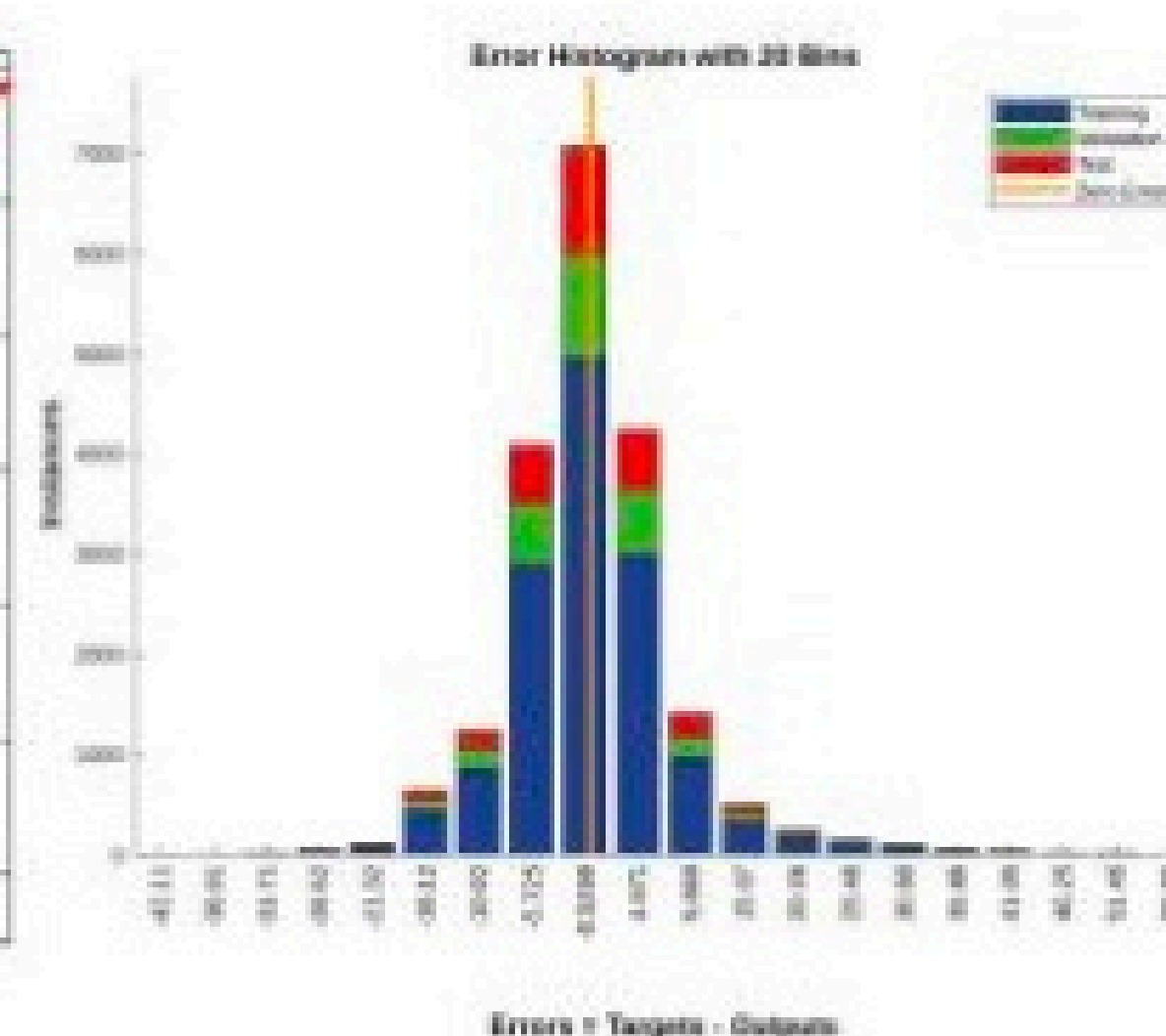


Figure 9: The error histogram of the neural network in all 3 stages: training, validation, and testing

Future Work:

The plan for the future is to hopefully implement multi-layered Neural networks in order to get a near-perfect ranking. To hopefully integrate smart technology like Digital Twin to be able to run simulations on the optimal factors that we have to get more accurate data on how their future will be, how they'll work, and the possible problems they might face in the future in order for us to get more accurate data on the factory and which ones are suitable for real-life implementation.

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